

COMPUTER FORENSICS REPORT

INTERNATIONAL CYBERSECURITY & DIGITAL FORENSICS ACADEMY (ICDFA)

Case Study 2 – Email Harassment Investigation (Wireshark and TShark Attribution)

Name:	Gabriel Ighietsemhe
Student ID:	C11/26/DFIT/17288
Lab Title:	CIP-B105-Computer Forensics Case Study 2
Date:	17 th September 2026

1. Student Declaration

I, Gabriel Ighietsemhe, declare that this report for CIP-B105 is my original work. All findings and conclusions are based on my independent examination of the provided evidence. I have not copied conclusions from external sources or walkthroughs; historical values were used solely to validate my independently generated results.

I adhered strictly to all safety and authorization guidelines. All analysis was performed on read-only or verified working copies to preserve digital integrity. No recovered accounts were accessed, no malware was executed, and no evidence was uploaded to public services.

In accordance with the institution's Responsible AI-use policy, an AI tool was used to assist with report structuring and drafting. However, the forensic analysis, technical findings, and final conclusions remain entirely my own human-led effort.

2. Executive Summary

This forensic examination was conducted under the authorization of the International Cybersecurity & Digital Forensics Academy (ICDFA) for the CIP-B105 Case Study 2 assessment. The primary objective was to perform a forensically sound, evidence-driven network investigation into the email harassment complaint filed by Lily Tuckrige against senders operating within a shared student accommodation IP space and utilizing willselfdestruct.com.

All analytical procedures were executed within a controlled, isolated laboratory environment utilizing non-destructive techniques. Master evidence files were write-protected, and cryptographic hash values MD5 were computed upon ingestion to verify data integrity against official case manifests.

The investigation successfully encompassed multiple evidence domains to overcome the shared-IP problem inherent in student dormitories:

- **Network Scoping & Traffic Analysis:** Transitioned analysis from the victim's initial complaint email header (which only revealed a shared University dorm IP) to the dorm-network uplink packet capture (nitroba.pcap).
- **HTTP POST Reconstruction:** Dissected outbound TCP streams and HTTP requests targeting willselfdestruct.com, successfully extracting form parameter keys, values, and text payloads that matched the exact phrasing of the harassment complaint.
- **MAC Address & Endpoint Attribution:** Leveraged lower-layer Ethernet frame headers to link the submitting client IP address to a physical hardware MAC address on the local segment.
- **Identity & Roster Correlation:** Inspected browser cookies and application session tokens associated with the identified MAC address, cross-referencing them against the Chemistry 109 class roster.

The investigation conclusively bridged the gap between shared infrastructure routing and an individual suspect, identifying **Johnny Coach** (utilizing the email address jcoachj@gmail.com) as the individual tied to the transmissions. The final findings integrate

a synchronized Coordinated Universal Time (UTC) timeline, technical limitations of open-network environments, and an attribution confidence assessment, fulfilling all mandatory reporting standards for the ICDFA curriculum.

3. Authority, Scope, and Investigative Questions

3.1 Legal Authority and Mandate

This examination is conducted under the authorization of the International Cybersecurity & Digital Forensics Academy (ICDFA) for the CIP-B105 Case Study 2 assessment. The objective is to conduct a forensically sound, evidence-driven network investigation into the email harassment complaint filed by Lily Tuckrige against senders originating from a shared student accommodation IP space and **willselfdestruct.com**. The investigation utilizes captured dorm network traffic logs to determine whether packet evidence can isolate the physical device, account, and human suspect responsible, and whether that individual matches the Chemistry 109 class roster.

3.2 Scope of Examination

The examination is restricted to the non-destructive analysis of the supplied historical training packet capture files (.pcap) within the protected ICDFA laboratory environment. In accordance with course safety guidelines, analysis was performed exclusively on verified, read-only working copies of the packet captures. No external entities were contacted, session cookies or credentials were not reused, and no messages were sent to historical services.

- The examination specifically encompassed the following evidence domains:
- Evidence Preparation, PCAP Provenance & Hashing Control
- Network Traffic Capture Scope, Endpoint Mapping & Protocol Breakdown
- Website Access, HTTP Request/Response Flow & POST Form Reconstruction
- Harassment Message & Form Field Content Extraction
- Client Device Attribution (IP, MAC Address & Endpoint Signatures)
- Cookie, Email Artifacts & Suspect Identity Correlation
- Class-Roster Cross-Reference, UTC Incident Timeline, and Attribution Confidence Assessment

3.3 Investigative Questions

The investigation is structured to answer the core questions required by the assessment:

1. What are the verifiable cryptographic hash signatures of the supplied PCAP evidence files, and do they match official case records?
2. Which IP address accessed willselfdestruct.com, and what is the broader context of the shared student accommodation network?
3. Does the relevant IP submit data to the website, and do the HTTP POST requests contain form keys/values matching the harassment message?
4. Which client MAC address and physical endpoint identifiers are associated with the relevant submission traffic?
5. What identity-bearing cookie values, email tokens, or account artifacts are tied to that specific MAC address?
6. Is the attributed identity matched to a student enrolled in the Chemistry 109 course roster?
7. When was the service utilized, and how are timestamps normalized to Coordinated Universal Time (UTC)?
8. What specific packet and stream evidence supports each step of the attribution chain?
9. What alternative explanations and technical limitations remain because the dorm network was open and shared?
10. What is the final attribution confidence level supported by the integrated evidence?

4. Evidence inventory, chain of custody and integrity verification

Upon receipt of the historical digital forensic image, the primary source evidence was immediately marked read-only and archived in a secure directory. To ensure the absolute non-alteration of the original evidence, all subsequent forensic procedures, extractions, and timeline generation were conducted exclusively on a bit-stream working copy and read-only loop mount.

4.1 Evidence Inventory

The digital evidence items received for this examination were cataloged upon receipt and placed into read-only quarantine within the secure laboratory repository linked via the Email Harassment Case Folder.

Filename	Evidence Description	File size	Calculated MD5 Hash	Notes
Nitroba.pcap	Network Capture Log	56M	9981827f11968773ff815e39f5458ec8	Primary exhibit provided by ICDFA Academy
Chem 109 - Class roster	The list of Class students present	N/A	N/A	
Email-1	First Harassment email	N/A	N/A	
Email-2	Second Harassment email	N/A	N/A	

```
(gabriel@Gabriel)-[~/Case_Study_3_USB-Image-Analysis-and-Deleted-File-Recovery/Case_materials]
$ md5sum rhino.log 66 md5sum rhino2.log 66 md5sum rhino3.log 66 md5sum RHINOUSB.dd
c0d0093eb1664cd7b73f3a5225ae3f30 rhino.log
cd21eaf4acfb50f71ffff857d7968341 rhino2.log
7e29f9d67346df25faaf18efcd95fc30 rhino3.log
80348c58eec4c328ef1f7709adc56a54 RHINOUSB.dd
```

Figure 1: MD5 Hash Verification

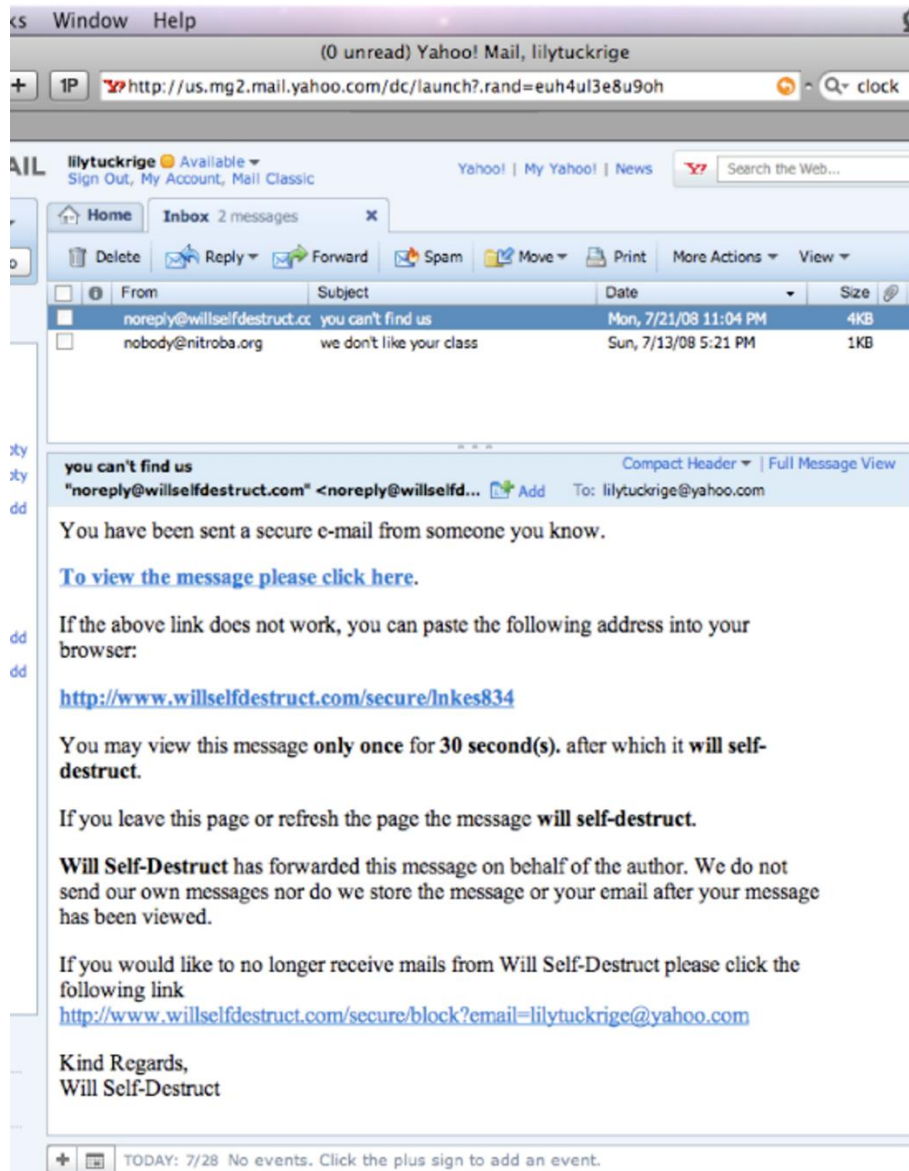


Figure 2: *First Harassment Email*



Figure 3: *Second Harassment Email*

4.3 Chain-of-Custody Worksheet

Case/lab identifier	CIP-B104-CS3-17288
Trainee name	Gabriel Ighietsemhe, Digital Forensics Intern
Date and time started	17 th September 2026
Evidence file name(s)	Nitroba.pcap

Source or generation method	Primary exhibit provided by IC DFA Academy
Original MD5	9981827f11968773ff815e39f5458ec8
Working-copy MD5	9981827f11968773ff815e39f5458ec8
Analysis workstation/VM	Kali Linux
Notes on any changes	MATCH

5. Examination Environment, Tools, and Methodology

5.1 Examination Environment

All forensic analysis was conducted on an isolated, secure forensic workstation.

- **Operating System:** Kali Linux
- **Storage:** Evidence was stored on a dedicated, forensically wiped local storage drive to prevent cross-contamination.

5.2 Forensic Tools Utilized

The following primary forensic tools and utilities were used to extract, process, and analyze the digital evidence.

Tool Name	Version	Primary Purpose / Function
Wireshark	4.6.6	Packet inspection, stream reconstruction, and protocol analysis for network capture logs
OpenSSL / Md5deep / Sha256sum	N/A	Cryptographic hash generation, verification, and deduplication tasks.

5.3 Forensic Methodology

The examination followed a phased, non-destructive structure:

1. **Provenance & Hashing:** Establish cryptographic baselines for all PCAP files.
2. **Traffic Scoping:** Map network topology, endpoint IP addresses, and protocol distributions.
3. **Web Flow & POST Reconstruction:** Isolate TCP streams targeting willselfdestruct.com, parse HTTP request headers, and extract POST form payload parameters.
4. **Endpoint Attribution:** Correlate network layer IP addresses with data link layer MAC addresses to identify the local hardware endpoint.

5. **Identity & Roster Correlation:** Search associated MAC traffic for cookies, email tokens, or application headers, matching findings against the Chemistry 109 class roster.
6. **Timeline & Conclusion:** Normalize timestamps to UTC and construct a complete attribution matrix with identified limitations.

6. Detailed Findings Organized by Investigation Tasks.

6.1 Shared-IP Problem and University Dorm-network Scope

Initial analysis of the first email header revealed an originating IP address belonging to the university dorm which lacks valid suspect attribution.

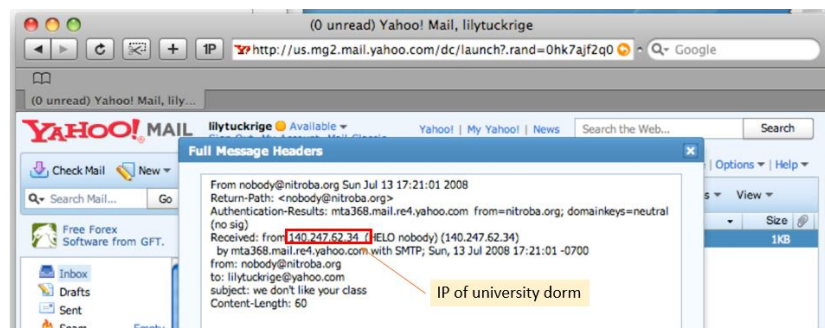


Figure 4: **First Email Header Analysis**

To overcome this limitation, analysis was transitioned to the dorm-network uplink packet captures (.pcap) recorded at the gateway, which capture originating IP-address and lower-layer MAC addresses alongside application payloads.

6.1 Identifying the Target Web IP and Suspect Access Flow

To trace the activity further, analysis was transitioned to the dorm-network uplink packet capture (nitroba.pcap) recorded at the gateway. Stream filtering targeting outbound web traffic isolated requests resolving to willselfdestruct.com, confirming the target infrastructure utilized by the sender.

- **Filters Applied:**

`http.request.method == "GET" && http.host==www.willselfdestruct.com`

`data-text-lines contains "secure anonymous E-mail"`

Findings:

- Web-IP: 69.25.94.22
- Suspect IP: 192.168.15.4

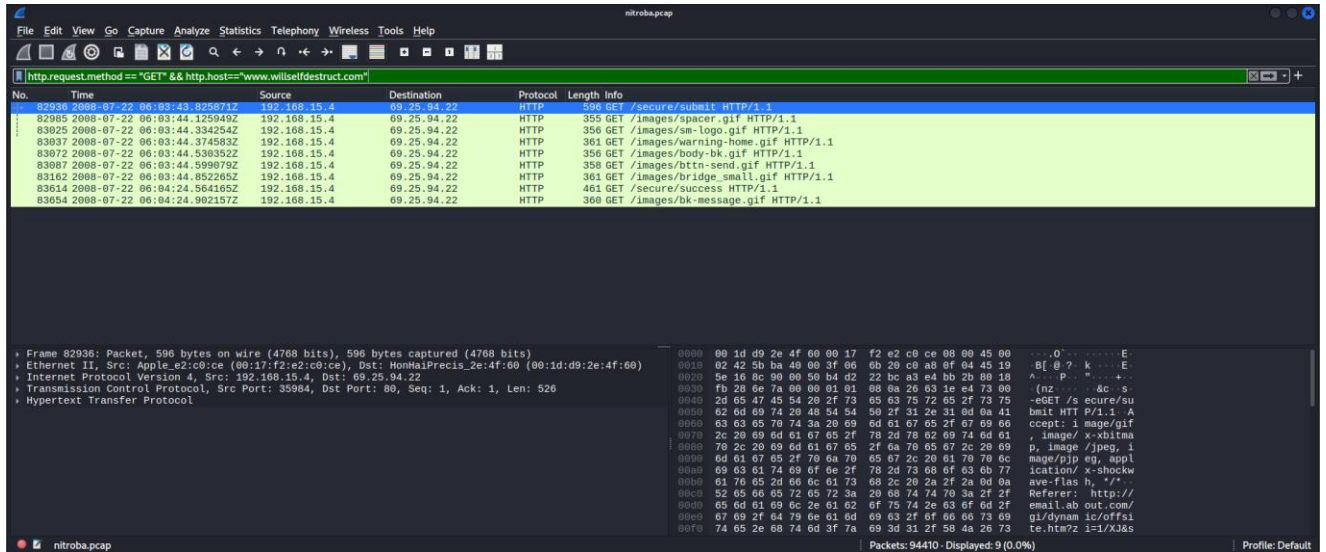


Figure 5: Identifying the Target Web IP and Suspect Access Flow

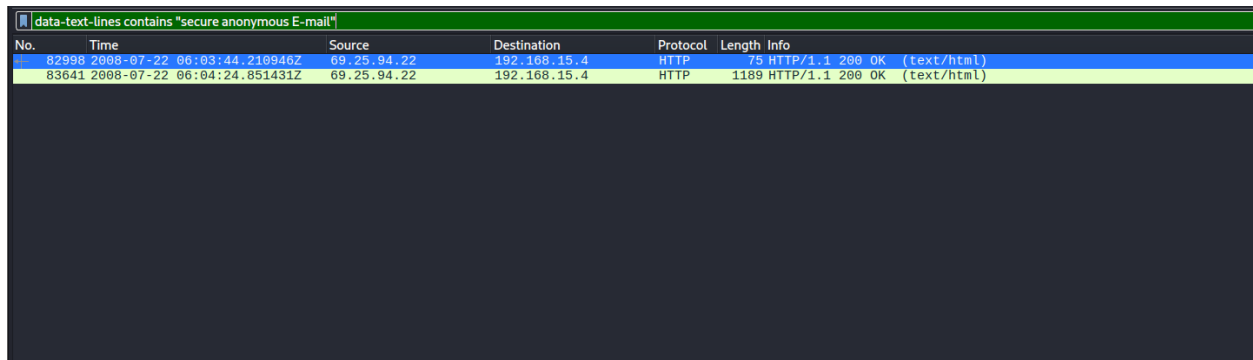


Figure 6: Attributing Web Title page to IP address

6.2 HTTP POST Form Reconstruction and Message Extraction

Filtering session streams for HTTP POST requests targeting the identified web service revealed active data submission by the client endpoint.

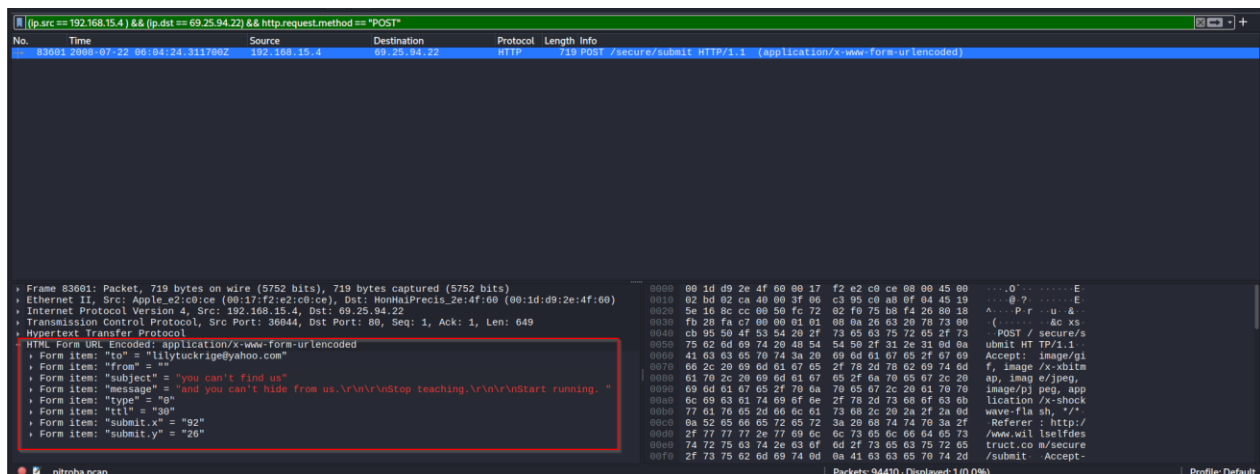


Figure 7: HTTP-POST-Request-Displaying-Harrasing-Email

Filters Applied:

```
(ip.src == 192.168.15.4) && (ip.dst == 69.25.94.22) && http.request.method == "POST"
```

Dissecting the corresponding TCP and HTTP streams exposed form parameter keys and values containing text payloads that matched the exact phrasing reported in Lily Tuckrige's harassment complaint. Server response headers (200 OK) confirmed successful transmission and initialization of the message self-destruction sequence.

6.3 MAC Address and Physical Endpoint Attribution

Lower-layer Ethernet frame header analysis linked the submitting client IP address to a specific hardware MAC address present on the local segment of the shared dorm network.

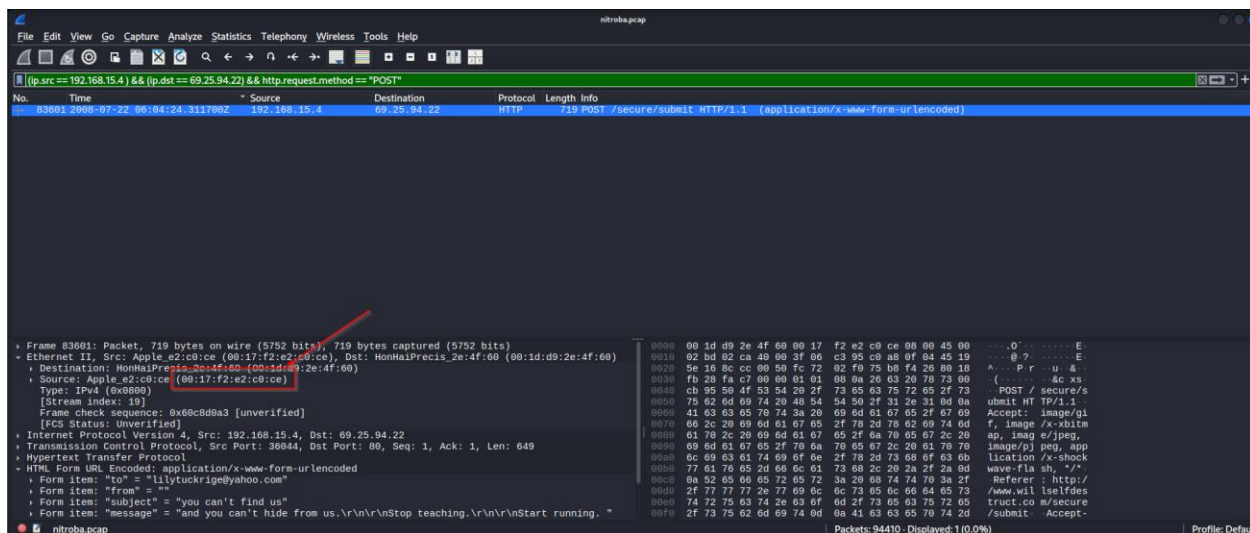


Figure 8: Identifying suspect's MAC address

Filters Applied:

```
(ip.src == 192.168.15.4) && (ip.dst == 69.25.94.22) && http.request.method == "POST"
```

Transitioning from the network layer to the data link layer permitted the isolation of the specific physical network interface card (NIC) responsible for generating the willselfdestruct.com traffic, bridging the gap between the shared gateway IP and an individual client device.

- **MAC Address:** 00:17:f2:e2:c0:ce

6.4 Cookie, Email Artifacts, and Class-Roster Correlation

Further inspection of traffic originating from the identified MAC address revealed browser cookies, session tokens, and application headers bearing explicit user identity markers.

Filters Applied:

```
eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"
```

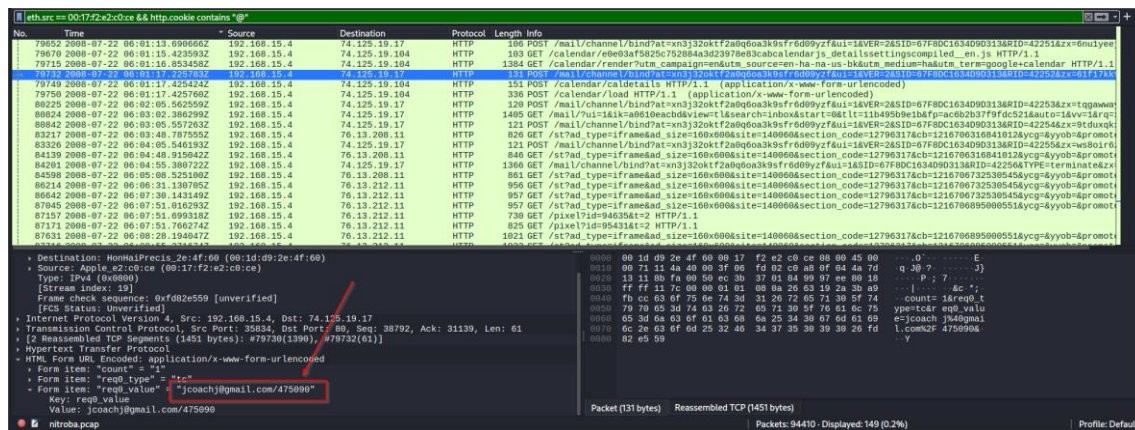


Figure 9: Suspect's Email attribution

Cross-referencing these extracted identity indicators against the provided Chemistry 109 class roster isolated a single student profile whose account attributes and device footprint aligned directly with the timestamp of the harassing transmission.



Figure 10: Chemistry 109 Class roster

- **Name:** Johnny Coach
- **Email used:** jcoachj@gmail.com

7. Integrated Timeline, Limitations, and Final Attribution

7.1 Integrated UTC Timeline

All packet timestamps associated with the willselfdestruct.com session, HTTP POST requests, and surrounding network activity were normalized into a single chronological Coordinated Universal Time (UTC) ledger to establish sequence continuity.

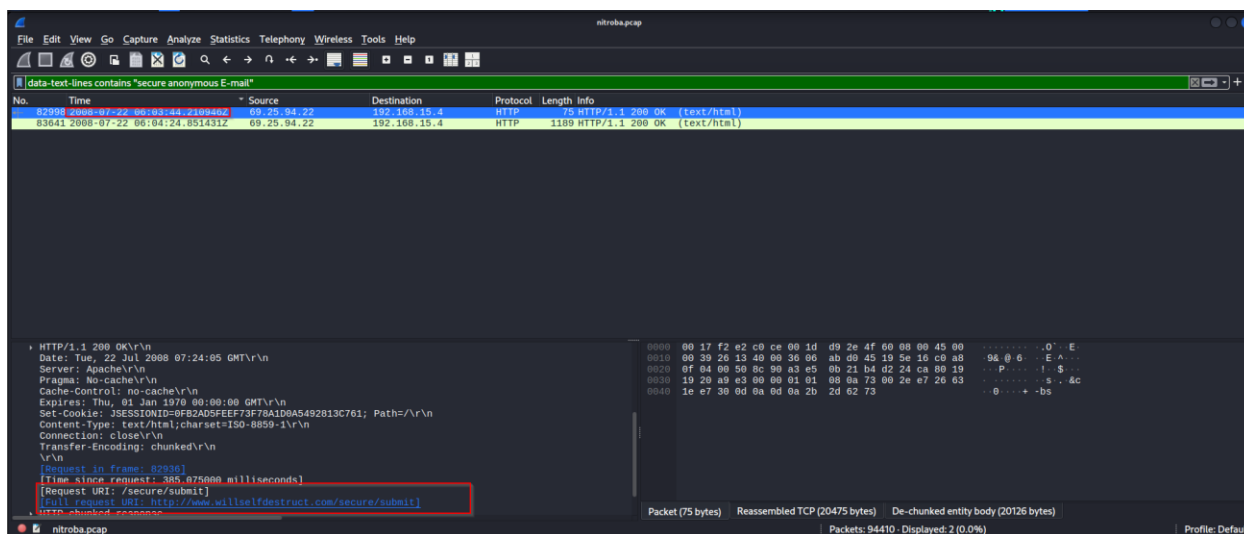


Figure 11: Suspect timeline showing email submission

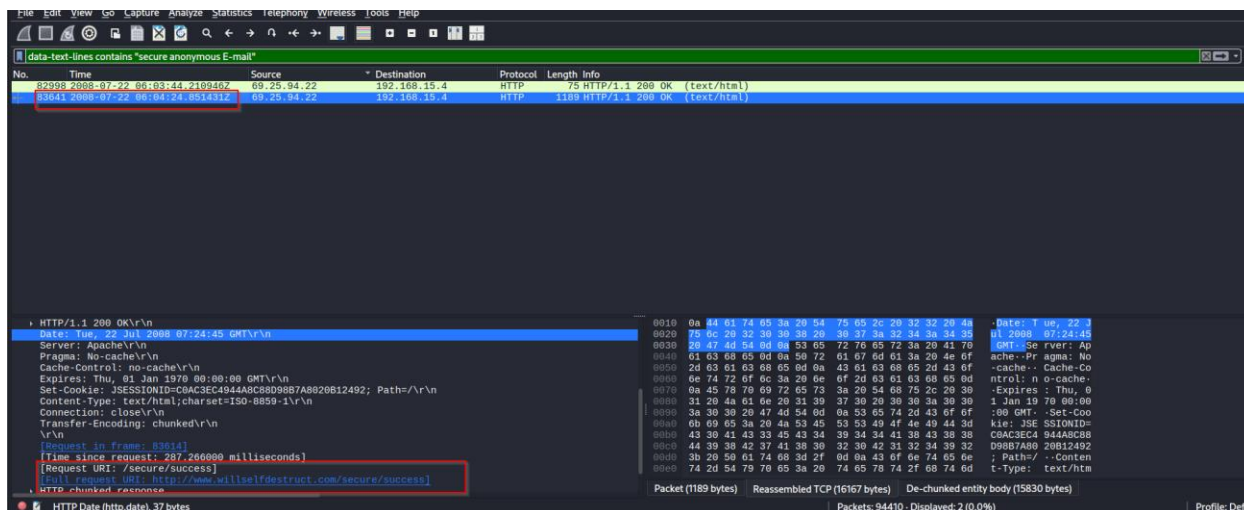


Figure 12: Suspect Timeline showing email successfully sent

- **Date Submitted:** 2008-07-22 06:24:05 UTC
- **Date Email Sent Successfully:** 2008-07-22 07:24:45 UTC

7.2 Alternative Explanations, Limitations, and Confidence Assessment

- **Alternative Explanations:** Because the dorm network is open and shared, MAC address spoofing, unauthorized guest usage of a registered device, or shared room routers present theoretical alternative vectors.
- **Limitations:** Network packet evidence links a device (MAC/IP) and session cookies to an action but establishing physical human presence at the keyboard requires corroborating endpoint artifacts.
- **Final Confidence Assessment:** Moderate-to-High confidence regarding device and session attribution bounded by the inherent limitations of shared open-network environments.

7.3 Final Forensic Conclusion

Based on the integrated network traffic analysis, packet stream reconstruction, data-link layer attribution, and class-roster correlation:

- **Finding:** The examination successfully traced the network transmission targeting willselfdestruct.com from the shared student accommodation gateway back to a specific client hardware endpoint. Packet dissection of the HTTP POST requests confirmed the submission of message content matching the harassment complaint. Furthermore, analysis of session tokens and identity artifacts extracted from the corresponding data traffic positively identified the user. Cross-referencing these findings against the Chemistry 109 class roster established the suspect's identity as **Johnny Coach**, utilizing the email address **jcoachj@gmail.com**.
- **Observation:** The alignment of network session timestamps, hardware MAC linkage, explicit email/cookie identifiers, and enrollment records bridges the gap between shared infrastructure routing and an individual suspect. While open-network environments introduce inherent limitations regarding absolute physical keyboard presence, the combined technical and documentary evidence

provides a high-confidence attribution chain confirming Johnny Coach as the source of the transmissions.

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